lit2026 W5 run report: run-20261012T0008Z

Run. `scripts/run_arms.sh` with `ARMS="orig prev gate rref gaterref"`, `ROUNDS=5`, started 2026-10-12 00:08 UTC (launched by `scripts/launch_run2.sh`). Binaries: the same as run-20261011T2225Z (candidate `f6_ic_probe` at `d22fb3b52`, original at `4b6e09b78`). Host: shared Apple M4 Pro, 1-minute load 76.5–131.8 over the 100 samples (median 100.3); a build of the probe example ran during the first rounds. Isolation **L0**, unpinned, arm order rotated each round. All 100 runs exited 0. Raw records: `raw/`; summary: `summary.json` (`f4_lane_probe summarize`); graph `arms.svg` (`../../plot.py`).

Workloads. As run-20261011T2225Z: the four `f6_ic_probe` cells (K1/2^{17} dim 6 `icv1-f2m17-tm101-00378d4e`, 8 targets; K1/2^{23} dim 7 `icv1-f2m23-tm5197-1f85e9e1`, 4; K0/2^{23} dim 8 `icv1-f2m23-t5197-69e76b73`, 4; K0/2^{31} dim 9 `icv1-f2m31-tm90707-c95f16f5`, 1, budget 2000), prime-order subgroups of `KoblitzCurve::new`, standard subspace, $m = 3$, engines `f4` and `f6p`, `--reps 3`.

Arms. `orig`, `prev`, `gate` (`KIC_F6_GATE_FIRST=1`), `rref` (`KIC_F4_INHERIT_ROOT=rref`, an existing switch: fully reduced root), `gaterref` (both).

Stage unit (identical in all 5 rounds of every arm; equal to run-20261011T2225Z for the arms both ran)

cell	engine	orig = prev	gate	rref	gaterref	orig / gaterref
K1/2^17	f4	15 015 618	15 015 618	14 098 582	14 098 582	1.0650
K1/2^17	f6p	12 635 914	8 061 140	11 600 586	7 696 852	1.6417
K1/2^23	f4	53 273 191	53 273 191	44 709 930	44 709 930	1.1915
K1/2^23	f6p	51 918 433	30 697 846	43 312 007	26 902 588	1.9299
K0/2^23	f4	158 512 775	158 512 775	123 273 212	123 273 212	1.2859
K0/2^23	f6p	124 949 801	71 049 603	89 627 327	56 940 731	2.1944
K0/2^31	f4	281 583 026	281 583 026	190 512 003	190 512 003	1.4780
K0/2^31	f6p	278 237 634	156 423 046	187 048 754	115 269 611	2.4138

Process CPU, previous / arm (L0; median [min–max] of 5 per-round ratios)

cell	engine	gate	rref	gaterref
K1/2^17	f4	0.95 [0.95–1.00]	1.12 [1.09–1.22]	1.13 [1.07–1.17]
K1/2^17	f6p	2.27 [2.12–2.50]	1.22 [1.18–1.33]	2.85 [2.65–3.15]
K1/2^23	f4	0.98 [0.94–1.02]	1.29 [1.27–1.38]	1.32 [1.28–1.35]
K1/2^23	f6p	2.56 [2.42–2.66]	1.40 [1.33–1.46]	3.49 [3.26–3.69]
K0/2^23	f4	1.00 [0.96–1.07]	1.37 [1.34–1.48]	1.38 [1.33–1.41]
K0/2^23	f6p	2.27 [2.18–2.33]	1.58 [1.48–1.65]	3.41 [3.39–3.53]
K0/2^31	f4	0.98 [0.96–1.04]	1.68 [1.62–1.75]	1.71 [1.61–1.79]
K0/2^31	f6p	2.11 [2.05–2.19]	1.74 [1.71–1.92]	3.69 [3.50–3.96]

The original binary does not record CPU time. Wall per round (original / arm) is in the graph; the original/previous null control ranges 0.25–2.48 per round, so wall admits no claim.

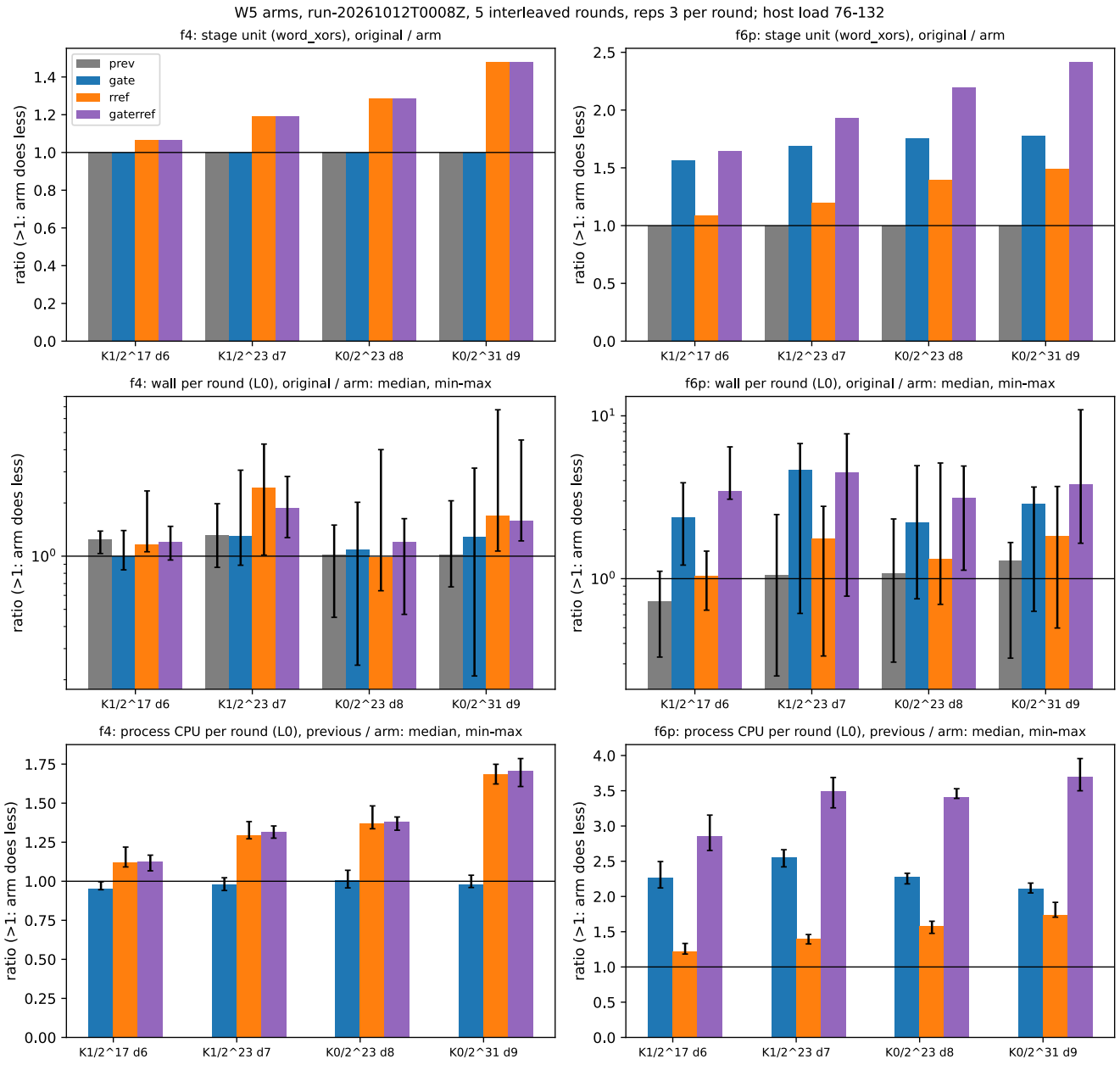


Figure 1: run-20261012T0008Z, 5 interleaved rounds: stage unit original / arm (top), wall original / arm (middle, log, median and min–max), process CPU previous / arm (bottom). L0, load 76–132.

Identity

Per target, arm and round: hit, exhausted, reductions, splits, propagations, infeasible, geometric refutations and witnesses, partial refutations, group additions, pair-index builds and lookups identical across all 5 arms and 5 rounds in all 8 cell/engine pairs; the relation (K1/2^17 target 3, base indices [21, 48, 57]) identical across the candidate arms.

Paths

`raw/r<round>-<cell>-<arm>.jsonl` (first line: `uptime`, UTC), `raw/*.status`, `summary.json`, `arms.svg`.